

SolarOrbz

Spherical Planet Generator — Reference Guide

An Unreal Engine plugin for building procedural and hand-authored spherical planets: a World Machine / World Creator-style layered terrain stack, biome placement driven by climate and shape, a whole-planet wind and moisture simulation, and thermal/hydraulic erosion.

Covers version 1.0 of the plugin.

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1. Overview & Architecture

SolarOrbz builds a planet in three ordered stages, each optional except the base sphere itself:

- **Terrain** — an ordered stack of layers (noise, heightmap, stamps, erosion) displaces the sphere radially. Layers composite top-to-bottom with Blend Modes and Weights, the same mental model as World Machine or World Creator.
- **Climate** — an optional whole-planet simulation computes real temperature (Kelvin) and moisture (wind-driven, with orographic rain shadows) on an independent grid, decoupled from mesh resolution.
- **Biomes** — an ordered stack of (Biome, Mask, Opacity) layers paints extra terrain detail and identifies a dominant biome per point, gated by masks that read elevation, slope, latitude, temperature, and moisture.

All three stages are separate, reusable Data Assets. A single Terrain Layer Stack, Climate Simulation, or Biome Stack can be authored once and referenced by many planet actors — e.g. one Terrain Stack for "Earthlike", one for "Mars", one for a barren asteroid, each swappable independently.

A key architectural point: most terrain layers are pure functions of a single point on the sphere (given a direction, return a height). Erosion and Climate are different — they need to know about the terrain around a point, not just the point itself, to simulate material or water moving across the surface. Both solve this the same way: they bake a coarse, independent lat/long grid once per regenerate (not once per vertex), then bilinear-sample that grid wherever a height or climate value is needed. This keeps their cost independent of mesh density.

2. Getting Started

There are two entry points into the plugin: the SolarOrbz dock tab (Window menu), which gives quick controls for the base sphere (Radius, Vertices Per Meter, Max Subdivisions) and a Generate / Bake / Clear workflow with no terrain authoring; and the full pipeline, which means placing a **SolarOrbz IcoSphere** actor directly in the level and configuring it via its Details panel. The full pipeline is what unlocks terrain, climate, and biomes.

A typical first planet:

- Place a **SolarOrbz IcoSphere** actor in the level.
- Content Browser → right-click → Miscellaneous → Data Asset → create a **SolarOrbzTerrainLayerStack**. Open it, add a **Planetary Noise Layer** to its Layers array (see Section 4.3).
- Assign that Terrain Layer Stack to the actor's **Terrain Stack** property.
- (Optional) Create a **SolarOrbzClimateSimulationAsset** and assign it to the actor's **Climate Simulation** property (Section 5).
- (Optional) Build a Biome Stack: create one or more **SolarOrbzBiome** assets, one or more **SolarOrbzBiomeMaskPreset** assets (each wrapping a Climate Mask), a **SolarOrbzBiomeStack** combining them, and assign it to the actor's **Biome Stack** property (Section 6).
- Press the actor's **Regenerate Mesh** button in its Details panel.
- When happy with the result, press **Bake To Static Mesh** to write it out as a permanent UStaticMesh asset at the configured Bake Package Path / Bake Asset Name.

Read this before you start tweaking numbers — it will save you real time:

Editing a referenced Data Asset does not automatically regenerate any actor using it. Terrain Layer Stacks, Climate Simulation assets, and Biome Stacks are all separate assets from the actor. Changing a slider inside one of

them (e.g. a Noise Layer's Amplitude, opened via its own asset editor) does not trigger the actor's auto-regenerate — only changing the actor's own properties does that. After editing any referenced asset, go back to the actor and press **Regenerate Mesh** manually. This is the single most common source of "I changed a value and nothing happened".

3. The IcoSphere Actor

ASolarOrbzIcoSphereActor is the actor you place in the level. It owns the **ProceduralMeshComponent** preview and orchestrates Terrain, Climate, and Biome evaluation every time it regenerates.

Property	Type / Units	Default	Description
Radius Meters	float, meters	1000	Sphere radius. No upper limit — performance is the only ceiling at extreme scale/density.
Vertices Per Meter	float	1.0	Desired vertex density along the surface.
Max Subdivisions	int	6	Hard cap on icosphere subdivision level. If Vertices Per Meter would need a higher level than this at the current Radius, a warning is logged and the density setting is not fully reached.
Terrain Stack	SolarOrbz Terrain Layer Stack asset	None	Procedural/authored terrain recipe. See Section 4.
Biome Stack	SolarOrbz Biome Stack asset	None	Biome placement and extra detail. See Section 6.
Climate Simulation	SolarOrbz Climate Simulation asset	None	Whole-planet temperature/moisture simulation. See Section 5.
Show Biome Debug Colors	bool	false	Colors each vertex by its dominant biome's Preview Color instead of the normal material. Needs Debug Biome Material set to an unlit material with Vertex Color wired to Emissive Color (see Section 8).
Debug Biome Material	Material Interface	None	The unlit/vertex-color material used when the above is on.
Default Material	Material Interface	None	Material restored when Show Biome Debug Colors is off. Leave unset to fall back to the engine default.
Enable Preview Collision	bool	false	Builds simple collision on the live preview mesh. Leave off for large/dense previews; add cheaply to the baked mesh instead.
Bake Package Path	FString	/Game/SolarOrbz/Meshes	Content Browser folder the baked mesh is written to.
Bake Asset Name	FString	SM_IcoSphere	Asset name for the baked static mesh.

Regenerate Mesh (button) rebuilds the preview from current parameters; also runs automatically whenever one of the actor's own properties above changes. **Bake To Static Mesh** (button) writes the current preview out to a permanent asset.

4. Terrain Layers

A **Terrain Layer Stack** (`SolarOrbzTerrainLayerStack`) is an ordered array of layers. Each layer contributes a height (in effect, centimeters internally, always authored in meters on screen) at a given point on the unit sphere; the stack combines them top-to-bottom using each layer's Blend Mode.

4.1 Shared Layer Properties

Every layer type below inherits these three properties:

Property	Type	Default	Description
Enabled	bool	true	Disabled layers are skipped entirely.
Blend Mode	enum	Add	Add / Subtract / Multiply / Max / Min / Replace, combined against everything accumulated from layers below it in the stack.
Weight	float	1.0	Multiplies this layer's raw output before blending — the simplest way to fade a layer in or out without changing its other settings.

4.2 Noise Layer

Fractal (multi-octave) Perlin noise scaled by a raw amplitude in meters. Scale-agnostic by design — use this for small or irregular bodies (asteroids, tiny moons) where "sea level" isn't a meaningful concept. For planet-scale terrain, prefer the Planetary Noise Layer below.

Property	Type / Units	Default	Description
Seed	int	0	Randomizes the pattern without changing its statistical character.
Octaves	int, 1-10	5	Fractal octaves summed together. More = finer detail, more cost.
Frequency	float	2.0	Roughly "noise cycles per trip around the sphere". 0.5-3 = continent scale; 20+ = fine roughness.
Lacunarity	float, ≥ 1	2.0	Frequency multiplier applied each octave.
Persistence	float, 0-1	0.5	Amplitude multiplier applied each octave; lower values make finer octaves contribute less.
Amplitude Meters	float, meters	5	Peak-to-peak height range the fractal sum maps onto. Clamped 0..1,000,000.
Warp Strength	float	0.0	Domain warp strength — pushes the sample position through a second noise field first, breaking up regular noise patterning. 0 disables it.
Warp Frequency	float	1.0	Frequency of the warp field. Only used when Warp Strength > 0.

4.3 Planetary Noise Layer

The same fractal noise machinery as Noise Layer, but scaled by real meters relative to sea level instead of a raw amplitude. "Sea level" here means the planet's base radius — the same zero point Climate Simulation's Sea Level measures from. The value means the same thing whether the planet is Earth-scale or a 50m test sphere, so changing Radius never requires re-tuning this layer.

Property	Type / Units	Default	Description
Seed / Octaves / Frequency / Lacunarity / Persistence / Warp Strength / Warp Frequency	—	—	Identical to Noise Layer above (shared base class).
Max Elevation Meters	float, meters	8000	How high above sea level the tallest peaks reach. Clamped 0..200,000. Reference: Everest is ~8,850m.
Max Depth Meters	float, meters (positive = depth)	4000	How far below sea level the lowest points reach, entered as a positive depth. Clamped 0..200,000. Reference: Mariana Trench ~10,900m; average ocean depth ~3,700m.

Realistic elevation is proportionally tiny next to a planet's radius — this is physically correct (real Earth looks almost perfectly smooth from orbit despite Everest existing), not a bug. If a planet looks flat at default values, zoom the camera in close to the surface before assuming something is broken.

4.4 Heightmap Layer

Samples an authored equirectangular (lat/long) grayscale heightmap texture using the mesh's spherical UV — e.g. a real-world DEM for Earth or Mars. Editor-only: reads the texture's source pixel data directly so it works regardless of compression settings. Import with compression set to None so full precision survives.

Property	Type / Units	Default	Description
Heightmap Texture	Texture2D	None	X = longitude, Y = latitude, north pole at top. 16-bit or float source recommended for real DEMs.
Min Height Meters	float, meters	-11000	World-space height that black (0.0) in the heightmap maps to. Clamped $\pm 200,000$.
Max Height Meters	float, meters	8848	World-space height that white (1.0) in the heightmap maps to. Clamped $\pm 200,000$.

4.5 Stamp Layer

Places a single feature — a heightmap texture, or a procedural dome/crater — at a specific point on the sphere, with an angular radius and soft edge falloff. Mirrors World Creator's terrain stamping: use several of these for individual mountain ranges, craters, or impact basins instead of covering the whole globe with noise.

Property	Type / Units	Default	Description
Latitude / Longitude	float, degrees	0 / 0	Where the stamp is centered. Latitude -90..90 (south..north pole); Longitude -180..180.
Angular Radius	float, degrees	15	How far the stamp reaches from its center, as arc along the surface.
Edge Falloff	float, 0-1	0.25	Fraction of Angular Radius over which the stamp fades at its edge, so it blends instead of cliffing off.
Stamp Rotation Degrees	float	0	Rotates the stamp around its own center. Only matters with a heightmap assigned.

Property	Type / Units	Default	Description
Stamp Heightmap	Texture2D	None	Optional grayscale heightmap sampled across the stamp's footprint (orthographic projection — no seam concerns). Leave unset for a procedural dome/crater instead.
Amplitude Meters	float, meters	10	Peak height — the dome/crater's extremum, or the heightmap's white value. Clamped 0..1,000,000.
Crater	bool	false	Only used without a heightmap: carves a crater (dips at the center, optional rim) instead of a dome.
Crater Rim Height	float	0.3	Only used with Crater: raised rim at the crater's edge, as a fraction of Amplitude Meters. 0 disables the rim.

4.6 Erosion Layer

Unlike every other layer, Erosion needs the combined height of every layer below it across the *whole planet*, not just a single point, to simulate material moving downhill. It bakes a delta-height grid once per regenerate and bilinear-samples that grid afterward. Place it after your Noise/Heightmap layers so it has real terrain to erode; leave Blend Mode at the default Add, since its own output is a height delta.

Two passes, matching classic terrain-erosion tooling:

- **Thermal erosion** — loose material above a talus angle slides downhill each iteration until slopes stabilize. Turns sharp noise-generated cliffs into believable scree slopes.
- **Hydraulic erosion** — water is routed downhill (steepest-descent, 8-neighbor) across the whole grid each pass, carving where flow is strong and depositing where it stalls. This is a deterministic grid-based flow-routing model, not particle/droplet-based, so a full spherical bake is one pass rather than tracking thousands of random-walking droplets.

Property	Type / Units	Default	Description
Grid Width / Grid Height	int	256 / 128	Bake grid resolution. Independent of mesh density — higher gives finer drainage detail at the cost of a slower bake.
Enable Thermal Erosion	bool	true	Toggles the thermal pass.
Talus Angle Degrees	float, 1-89	33	Maximum stable slope. Loose scree/sand settles ~30-35°; bare rock holds much steeper. Anything steeper redistributes toward the lowest neighbor.
Thermal Iterations	int	20	Relaxation passes. Most visible change happens in the first several.
Thermal Strength	float, 0-1	0.5	Fraction of the excess-over-talus height moved per iteration. Low = gradual; high = aggressive, can overshoot.
Enable Hydraulic Erosion	bool	true	Toggles the hydraulic pass.
Hydraulic Iterations	int	4	Full downhill-routing passes. Each can deepen channels and let water re-route across reshaped terrain.
Rainfall Amount	float	0.02	Water added at every cell at the start of each pass. Uniform across the whole planet — not yet driven by a Climate Simulation's real rainfall.
Erosion Rate	float	0.3	How readily flowing water carves material. Higher cuts deeper channels for the same flow and slope.
Deposition Rate	float	0.3	How readily water drops carried sediment once it can no longer carry it (e.g. reaching a basin).

Slope/talus math uses a single equatorial cell-spacing estimate uniformly across the grid, rather than correcting for longitude cells shrinking toward the poles — erosion is therefore slightly less physically accurate near the poles than the equator.

5. Climate Simulation

A **SolarOrbzClimateSimulationAsset** runs once per regenerate, right after the Terrain Stack displaces the mesh, on its own lat/long grid (independent of mesh resolution, sampling elevation from the same Terrain Stack the mesh uses). It produces two fields for every point on the planet, consumed by Climate Masks (Section 6.3):

- **Temperature** (Kelvin, not normalized) — a latitude baseline minus an elevation lapse rate. A real physical unit, so an icy moon and a Venus-hot furnace are both directly representable, not just "hotter/colder than Earth".
- **Moisture** (0 driest .. 1 wettest) — wind is advected around each latitude band, picking up moisture over ocean cells and raining it out over land, with extra rainfall on uphill slopes (the orographic effect that produces real rain shadows behind mountain ranges). Scaled by sea-level atmospheric density, so a thin atmosphere runs drier and a thick one runs wetter for the same wind tuning.

Property	Type / Units	Default	Description
Grid Width / Grid Height	int	256 / 128	Simulation grid resolution. Independent of mesh density.
Equator Temperature	float, Kelvin	288	Temperature at the equator, at sea level. Earth ~288K. Not clamped to any Earth-relative range.
Pole Temperature	float, Kelvin	255	Temperature at the poles, at sea level. Earth ~255K.
Lapse Rate Per Km	float, K/km	6.5	Kelvin dropped per km of elevation above sea level. Earth's environmental lapse rate is ~6.5 K/km.
Sea Level	float, meters	0	Elevation below which a cell counts as ocean (a moisture source and warmth-moderated zone). Clamped $\pm 200,000$.
Atmosphere Density At Sea Level	float, kg/m ³	1.225	Earth ~1.225; Mars ~0.02 (thin); Venus ~65 (thick). Scales Moisture Capacity and Evaporation Rate below (square-rooted, so the huge real-world range doesn't become an unusable multiplier). A deliberate simplification, not a full atmospheric model.
Trade Wind Edge Degrees	float, 0-90	30	Latitude where tropical easterlies give way to mid-latitude westerlies.
Westerlies Edge Degrees	float, 0-90	60	Latitude where mid-latitude westerlies give way to polar easterlies.
Reverse All Bands	bool	false	Flips every wind band's direction at once — for a planet with retrograde rotation.
Wind Loops	int, ≥ 1	2	Full circuits the wind simulation makes around each latitude band before the final circuit is recorded, letting carried moisture settle into a steady state.
Moisture Capacity	float	1.0	Maximum moisture the simulated air can carry, before Atmosphere Density scaling.
Evaporation Rate	float	0.15	Moisture gained per grid step over an ocean cell, before Atmosphere Density scaling.
Base Rainfall Rate	float, 0-1	0.05	Fraction of carried moisture that always rains out over land each step, even on flat ground.
Orographic Rainfall Factor	float	2.0	Extra rainfall per km of uphill elevation gain along the wind direction — produces rain shadows.

The wind model is a simplified three-cell circulation (tropical easterlies / mid-latitude westerlies / polar easterlies, mirrored across both hemispheres) — the same pattern that gives Earth its trade winds and prevailing westerlies, but it does not model real Hadley/Ferrel cell dynamics, seasons, or axial tilt.

6. Biomes

The biome system layers extra terrain detail and identifies a "dominant biome" at each point, using the same climate/shape context Climate Masks read from (elevation, slope, latitude, temperature, moisture).

6.1 Biome Assets

A **SolarOrbzBiome** is a reusable bundle: an identifying color, optional extra terrain detail, and placeholder scatter rules for a future PCG/PCGEx hookup.

Property	Type / Units	Default	Description
Preview Color	Linear Color	White	Identifier color used by mask previews and the Show Biome Debug Colors view.
Terrain Detail	SolarOrbz Terrain Layer Stack	None	Extra layers specific to this biome (e.g. a "Mountain" biome's own ruggedness noise), blended on top of the planet's base terrain wherever this biome's mask is active.
Scatter Entries	array of scatter rules	empty	Placeholder for a future PCG/PCGEx hookup: each entry has a Tag, Density Per Square Meter, and a Min/Max Slope range.

6.2 Masks & Mask Presets

A **SolarOrbzBiomeMaskPreset** is a standalone, reusable asset holding one mask (Climate Mask or Composite Mask). Create it directly in the Content Browser (right-click → Miscellaneous → Data Asset → SolarOrbzBiomeMaskPreset) and reference the same preset from as many biome layers or composite masks as you like — e.g. author "Near Poles" once and reuse it everywhere.

6.3 Climate Mask

The main classifier: a Whittaker-diagram-style combination of elevation, latitude/temperature, slope, and moisture, each expressed as a soft-edged **Mask Range** (Min, Max, Falloff) that returns 1.0 inside the range, ramping to 0.0 over Falloff at each edge. A disabled range always returns 1.0 (no restriction on that axis). All enabled ranges multiply together.

Axis	Type / Units	Notes
Elevation	Mask Range, meters	Relative to the planet's base radius.
Latitude	Mask Range, 0-1	0 = equator, 1 = pole. Purely geometric, no elevation. Prefer Temperature once a Climate Simulation is assigned.
Temperature	Mask Range, Kelvin	Simulated when a Climate Simulation is assigned; otherwise falls back to a generic Earth-like lerp by latitude (288K equator .. 255K pole).
Slope	Mask Range, 0-1	0 = flat ground, 1 = vertical cliff face.
Moisture	Mask Range, 0-1	From the wind/orographic simulation when a Climate Simulation is assigned; otherwise falls back to a low-frequency noise field (tuned by Moisture Seed / Moisture Frequency, only used in this fallback case).

6.4 Composite Mask

Combines several Mask Preset references so compound biome variants (e.g. "Icy Chemical Mountain") are built by stacking simple, reusable masks rather than needing a dedicated class or enum entry per combination.

Property	Type	Default	Description
Masks	array of Mask Preset refs	empty	The building-block masks to combine.
Combine Mode	enum	Multiply	Multiply = intersection ("all of these must apply" — typical for compound biomes). Also available: Min, Max (union — "any of these is enough"), Average.

6.5 Biome Stack

A **SolarOrbzBiomeStack** is an ordered list of (Biome, Mask, Opacity) layers, painted onto the planet Photoshop-style — the first entry is the bottom of the stack, the last is topmost / highest priority. At each point, the stack blends every layer's extra terrain height by that layer's mask weight × Opacity, and separately reports the single strongest matching biome (ties favor the topmost layer) — used by Show Biome Debug Colors.

Property	Type	Default	Description
Biome	SolarOrbzBiome asset	None	Which biome this layer paints.
Mask	Mask Preset asset	None	Where this biome applies. Leave unset for "always applies".
Opacity	float, 0-1	1.0	Overall strength multiplier for this layer, independent of the mask.

7. Units & Scale Conventions

- Every distance/elevation property you author is in **meters**. Internally the plugin converts to Unreal's native centimeters at the point of use — you never need to think in cm.
- "Sea level" consistently means the planet's base radius (elevation 0) across Planetary Noise Layer, Heightmap Layer, Climate Simulation's Sea Level, and Climate Mask's Elevation range — they all share the same zero point.
- Temperature is always absolute **Kelvin**, never a normalized 0-1 scalar, so genuinely non-Earthlike worlds (a lava planet, a cryogenic moon) are directly representable.
- Every elevation-related property has a generous but present upper clamp (typically 200,000m, i.e. 200km) enforced by Unreal on direct number entry, not just slider dragging — this exists specifically to catch an accidentally-mistyped extra zero before it produces a planet with mountains a thousand times taller than its own radius.
- Radius Meters and Vertices Per Meter on the actor itself were already meters-based and are unaffected by any of the above.

The Unreal Editor's own numeric-property display can still show a converted unit (commonly centimeters) regardless of what a property's underlying unit is, depending on a project-wide preference. If your sliders show "cm" instead of "m", go to **Edit** → **Project Settings** → **Editor** → **Appearance** → **Units** → **Advanced** and set **Distance / Length** to Meters. This is a project-wide editor preference, not a per-plugin setting — it will also affect distance display elsewhere in the project (actor transforms, static mesh dimensions, etc). The underlying values are correct either way; this only changes the display label.

8. Troubleshooting & Known Gotchas

"I changed a value and nothing happened."

Almost always the auto-regenerate gap described in Section 2: editing a Terrain Layer Stack, Climate Simulation, or Biome Stack asset directly does not automatically refresh any actor referencing it. Go back to the actor and press **Regenerate Mesh**.

"My planet looks like a perfectly smooth ball."

Check the Output Log (category **LogSolarOrbz**) after Regenerate Mesh — it logs the actual height range achieved each regenerate, as a percentage of the planet's radius. If that percentage is nonzero but small, this is very likely realistic-but-subtle scale (Section 4.3's note), not a bug — zoom in close to the surface to confirm. If it reads essentially zero, check that Terrain Stack is assigned, has at least one enabled layer with nonzero Weight, and that you pressed Regenerate Mesh after any asset edits.

"Show Biome Debug Colors is on but the mesh is black/gray."

Debug Biome Material must be an **unlit** material with a **Vertex Color** node wired directly to **Emissive Color**. The colors come from vertex data already baked into the mesh (each biome's Preview Color); the material's only job is to display it. A normal Lit material ignores vertex color and will render flat/washed-out even if everything else is correctly configured.

"Wind / atmosphere density settings on Climate Simulation don't seem to do anything."

Two likely causes, both diagnosable from the Output Log (category **LogSolarOrbzErosion** is separate; Climate logs under **LogSolarOrbz**, prefixed "SolarOrbz Climate:"): first, the auto-regenerate gap above — press Regenerate Mesh after editing the Climate Simulation asset. Second, if Moisture logs as completely flat, every grid cell is registering as ocean (elevation everywhere $\leq$ Sea Level) or the underlying terrain has no real elevation variation — wind and atmosphere density only affect how moisture behaves over land and mountains, so a featureless sphere gives them nothing to act on.

Logging categories

Category	What it reports
LogSolarOrbz	General regeneration flow: subdivision level actually reached vs. requested, terrain height range and what percentage of radius it represents, climate grid Temperature/Moisture min/max/average, which material got applied to slot 0, and warnings for common misconfigurations (missing Terrain Stack, missing Biome Stack with debug colors on, missing Debug Biome Material, zero biome-mask hits, flat moisture).
LogSolarOrbzErosion	Erosion bake stats each regenerate: grid size, which passes ran and for how many iterations, and the resulting delta-height range (deposited .. carved).

9. Known Simplifications

Worth knowing going in — none of these are bugs, but they're deliberate simplifications that shape how far the simulation-driven features can be pushed:

- **Hydraulic erosion** is a deterministic, grid-based flow-routing model (steepest-descent, processed highest-to-lowest each pass), not particle/droplet-based. Chosen so a full spherical bake is one pass instead of tracking thousands of random-walking droplets across a wrapped grid.
- **Erosion and Climate slope math** use a single equatorial cell-spacing estimate uniformly across their grids, rather than correcting for longitude cells shrinking toward the poles. Both are therefore slightly less physically accurate near the poles than the equator.
- **Wind** is a simplified three-band zonal circulation (tropical easterlies / mid-latitude westerlies / polar easterlies), not full Hadley/Ferrel/Polar cell dynamics, and has no concept of seasons or axial tilt.
- **Atmosphere Density's** effect on moisture capacity/evaporation is a deliberately simplified stand-in (a square-root scaling relative to Earth's sea-level density), not real atmospheric thermodynamics (vapor pressure, specific heat, etc).
- **Erosion's Rainfall Amount** is uniform across the whole planet by default, not yet driven by a Climate Simulation's actual computed rainfall — wiring the two together is a natural future extension.
- **Scatter Entries** on Biome assets are a placeholder schema (Tag, density, slope range) for a future PCG/PCGEx hookup — nothing currently reads or spawns from them.

End of reference guide. For the underlying C++ source, see the Source/SolarOrbz folder alongside this document.